

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API / Integration Path	Security / Scaling Note
Client Access and UI	React Web Application	User requests	HTTPS + REST API	Secure login using HTTPS
API / Backend Logic	Node.js + Express	Threat analysis requests	REST API	Input validation and authentication
Authentication and Identity	Firebase Authentication / JWT	User credentials	Authentication API	Token-based authentication
Database / Structured Records	MongoDB Atlas	Threat logs and risk scores	MongoDB Driver	Encrypted storage and backup
File / Object Storage	AWS S3 / Firebase Storage	Log files and reports	Storage API	Access control
Monitoring / Logs	AWS CloudWatch	Server logs and metrics	Monitoring API	Automatic alerts
Analytics / Reporting	React Dashboard	Risk reports and trends	REST API	Scalable visualization

Technical Justification

The SentinelX project uses cloud services to process cyber threat data efficiently. REST APIs connect the frontend with the backend, NLP analyzes threat intelligence, serverless functions calculate risk scores automatically, MongoDB Atlas stores data securely, and cloud monitoring provides scalability and reliability.